Comments on
Functional Inequalities on the Quantum Boolean Cube with Applications to Quantum Learning Theory

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Analysis on the dyadic group $\{-1,1\}^n$ (i.e. Boolean cube in the present paper) and its Banach-space-valued generalization have important applications in harmonic analysis, probability theory, and functional analysis. This paper is mainly concerned with developing its noncommutative theory. A natural noncommutative counterpart of dyadic group $\{-1,1\}^n$ is the so-called CAR algebra. The analysis on the CAR algebra goes back at least to [1], [CL93] and [LP98] . Another noncommutative counterpart of the dyadic group $\{-1,1\}^n$ is the matrix algebra $\mathbb{M}_{2^n} := \mathbb{M}_2^{\otimes n}$ (i.e. quantum Boolean cube). This paper focuses on the analytic theory on the matrix algebra (or the quantum Boolean cube) $\mathbb{M}_{2^n}$ and its applications.

I consider that the main contributions of this paper are as follows

(1) It is proved that the quantum Ornstein-Uhlenbeck semigroup is holomorphic on $L_p(\mathbb{M}_{2^n})$, $1 < p < \infty$. In the classical setting, this result was established by Pisier in his remarkable paper [Pi82a]. More precisely,

My suggestions are listed below

1.

References

[1] S. Ayupov, S. Ferleger, and F. Sukochev. Isomorphism between the associative and non-associative L_p-spaces of type II_1 hyperfinite factors. Mathematica Scandinavica (1996): 271-285.